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REVIEWS

The Probability Approach in Econometrics. By TRYGVE HAAVELMO. (University of Chicago : Cowles Commission Paper No. 4, 1944. Pp. viii + 118.)

THIS publication, which has also been issued as a supplement to *Econometrica*, should be read by everyone interested in statistical economics. Since the discussion between Keynes and Tinbergen in this JOURNAL on the logic of econometric research a number of articles have appeared with the object of expounding and justifying the procedures used in this field.¹ This paper is the most comprehensive exposition of all, and combines a clear statement of the logical foundations of econometrics with an account of the author's brilliant contribution to that subject, which is having such a revolutionary effect on the procedures used in the estimation of economic relationships. In particular, it may be recommended to those who are sceptical of econometric work, for, as Koopmans has remarked,² "the meaning of Haavelmo's work is that economic statistics has now caught up with economic theory. It has now become clear that methods of statistical estimation, even of a single equation of economic behaviour, must in some way take account of the fact that the variables entering that equation are part of the wider set of relevant economic variables which are determined by a complete system of simultaneous equations." The greater part of the exposition of this paper is intelligible to those without special mathematical equipment, despite the fact that mathematical notation is used freely and some parts of the argument are of a purely formal character. The analysis was first presented in mimeographed form in 1941, and the essentially new feature was presented in an earlier article in *Econometrica*.³

¹ In particular : "Econometric Business Cycle Research," by J. Tinbergen, in *The Review of Economic Studies*, Vol. VII, 1939-40, pp. 73-90; "The Logic of Econometric Business Cycle Research," by T. Koopmans, in *The Journal of Political Economy*, Vol. XLIX, April 1941, pp. 157-81; "Statistical Testing of Business-Cycle Theories," by T. Haavelmo, in *The Review of Economic Statistics*, Vol. XXV, February 1943, pp. 13-18; "Economic Interdependence and Statistical Analysis," by J. Marschak, in *Studies in Mathematical Economics and Econometrics* (1942), pp. 135-50.

² See "Statistical Estimation of Simultaneous Economic Relations," in *Journal of the American Statistical Association*, Vol. 40, December 1945, pp. 448-66.

³ See "The Statistical Implications of a System of Simultaneous Equations," in *Econometrica*, Vol. 11, January 1943, pp. 1-12.

The paper contains six chapters dealing respectively with : abstract models and reality ; the degree of permanence of economic laws ; stochastic schemes as a basis for econometrics ; the testing of hypotheses ; problems of estimation ; and problems of prediction.

An abstract model may be regarded simply as a restriction on the joint variations of a set of variables. It is pointed out in Chapter I that any exact model is likely to be contradicted by experience, so that one's model must be set up or one's theory formulated in terms of probability. In this less dogmatic form one's theory may be found not to be contradicted by observations. The observations must, however, be appropriate, so that with every theory we ought to specify what observable variables are to be identified with those in the theory, and also what procedures are admissible for carrying out the observations. This matter of specification is termed by Haavelmo a " design of experiments." If, for example, we suppose demand curves to be irreversible and to depend on the present position of the individual, then the relevant data could be obtained in principle by interviewing, but could not be obtained, even under the most favourable conditions, from historical records of prices and quantities. An irreversible scheme might, of course, be reduced to a reversible one by the introduction of additional variables.

Chapter II contains a discussion of the stability of economic relationships. The author points out that in many cases experience shows that relationships which are successful in explaining the variation of some economic phenomenon over one period break down when applied to a subsequent period, while, on the other hand, observations frequently satisfy much simpler relationships than are expected on theoretical grounds. An important problem in obtaining stable relationships arises from the difficulty of ensuring that the condition *ceteris paribus* is fulfilled in view of the fact that our observations are obtained, not by experiment or by randomising the factors with which we are not concerned, but from the complicated interactions of the real world. This means that we may have to introduce a comparatively large number of variables into any expression in order to hold other things constant and obtain the net relationship between, say, price and quantity in a demand study, while at the same time the smallness of the sample of observations available and the importance of common factors affecting the variation of many economic variables may make it impracticable to work with equations with so many terms. If we, then, neglect some of the terms

because in the sample period they are not from a purely empirical point of view needed, we run the almost certain danger that we shall be caught out in the next period, when the omitted terms again require separate considerations.

Because the movement of economic variables is restricted by a set of simultaneous relationships, there may be little opportunity for the hypothetical free variation, with which we start out in economic theory, to manifest itself. At the same time it is important that we should begin with a theoretical structure of autonomous relationships if we are to gain much insight into the processes of economic change. By way of analogy, Haavelmo remarks that if we want to understand the working of a motor-car, we should study thermodynamics, the dynamics of friction, etc., because these subjects lead to highly autonomous relationships in that field of investigation, while we should not spend time investigating empirically such relationships as that between the speed of the car and the angle of the accelerator pedal, which possess very little autonomy. This is clearly sound, and, if acted on by applied economists, should go far to close the gap which has frequently existed in the past between empirical and theoretical studies.

At the beginning of Chapter III it is pointed out that it is hopeless to expect exact functional relationships to be fulfilled in economics, since this position is not strictly encountered even in such exact sciences as astronomy. Accordingly, a stochastic, as opposed to a functional, scheme is set out which is more appropriate to economic investigations. The essential feature of such a scheme is that it allows for disturbances in the equations connecting the variables. These may take the form of errors of observation which can be associated with particular variables and disturbances due to omissions and simplifications which can be associated only with a given equation. The latter type, which is inevitably present in economic investigations, gives rise to a more fundamental type of difficulty than do mere errors of measurement. In particular, they require a revision of the existing theory of estimation of the constants in the equations connecting the variables, and this is discussed in Chapter V. Disturbances also lead to two distinct types of derived exact equations. One type is the solution which is given if we abstract from disturbances; the other is the expected value of the solution if disturbances are admitted. Thus if we have a demand equation derived from budget data of the form

$$v = e^a + \epsilon v^b$$

where the disturbance is represented by ϵ , a random variable with mean zero and variance σ^2 , then if we assume there are no errors we have :

$$v^* = e^a V^b$$

On the other hand, the expected value of v given V , $E(v|V)$, is different from v^* being given by

$$E(v|V) = e^{a + \frac{1}{2}\sigma^2} V^b$$

so that the behaviour of the average family of given income V , $E(v|V)$, is not the same as that which would be observed, v^* if all the families acted alike without the differences introduced by the disturbance ϵ .

Chapter IV contains a brief description of the Neyman-Pearson theory of testing statistical hypotheses, which will be familiar to readers of the fundamental work of these two writers contained in the two issues of *Statistical Research Memoirs*, issued before the war by the Department of Statistics, University of London, University College. It is helpful in enabling the economist to see how a theory must be formulated if the statistician is to get profitably to work. It brings out clearly that the economist must specify the set of *a priori* admissible hypotheses before a test is constructed. The more this process of restriction can safely be carried out, the more hope there is of making valid inferences.

Chapter V follows out the implications for estimating the coefficients of economic relationships of the fact that each relation is part of, and influenced by, a system of stochastic equations. This means that there are factors operating to cause disturbances which are not explicitly introduced into our theory, and, further, that economic relationships do not exist on their own, but are part of a system of relationships. It is here that Haavelmo has made an important original contribution to the theory of estimation by showing that, except in special cases, unbiased estimates of the parameters in equations which form part of a system of the above kind cannot be obtained by considering each equation in isolation. This discovery opens up a new method of estimation which requires that the system of equations be written down in advance, even if we are only interested in one relationship, and considered as a whole. The difficulty of putting this principle into practice in the present state of knowledge calls for an examination of the circumstances in which the bias of the single-equation approach is especially serious, and of short cuts which can be adopted through the use of incomplete systems of two or more equations. These questions have been considered briefly in the

paper (1945) by Koopmans cited above, and will be dealt with more fully in a forthcoming publication of the Cowles Commission.¹

Chapter VI deals briefly with some of the problems of prediction. In his conclusion Haavelmo states as his belief that "if economics is to establish itself as a reputable quantitative science, many economists will have to revise their ideas as to the level of statistical theory and technique and the amount of tedious work that will be required, even for modest projects of research. On the other side we must count the time and work that might be saved by eliminating a good deal of planless and futile juggling with figures." This is my belief also, and I look forward to the day, already overdue, when it finds recognition in the syllabuses for degrees in economics.

In conclusion, attention may be drawn to work on another aspect of the estimation problem dealt with in Chapter V which is going on currently. In Haavelmo's scheme we concern ourselves with disturbances in the equations of a system, but neglect the sampling problems arising from errors of measurement in the variables. This latter question has been attacked recently by Tintner,² though in his case errors are dealt with to the neglect of disturbances. Haavelmo's estimation problem is, I think, novel in a way in which Tintner's is not, but both writers have put statistical economists very much in their debt.

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World Economic Survey, 1942-44. (Geneva: League of Nations (London: Allen & Unwin), 1945. Pp. 299. 10s.)

Commercial Policy in the Post-War World. (Geneva: League of Nations (London: Allen & Unwin), 1945. Pp. 124. 5s.)

THIS volume of the *Survey* covers the period from the autumn of 1942 to the close of 1944—a period of full war economy, when total world output was greater than ever before, although 50 million men were under arms. A general outline is followed by chapters on Production, Consumption and Rationing, Finance

¹ *Statistical Inference in Dynamic Economic Systems*, being the proceedings of a conference to be published in the series of Cowles Commission Monographs.

² See, in particular, "A note on Rank, Multicollinearity and Multiple Regression," in *Annals of Mathematical Statistics*, Vol. 16, September 1945, pp. 304-8, and "Multiple Regression for Systems of Equations," in *Econometrica*, Vol. 14, January 1946, pp. 5-36.